Double mass pendulum analysis

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When God plops a matrix in our laps we analyze it.

$$\begin{bmatrix}\ddot{u}_1(t)\\ \ddot{u}_2(t)\end{bmatrix} = \begin{bmatrix}-\frac{g}{r}-\frac{k}{m} & \frac{k}{m}\\ \frac{k}{m} & -\frac{g}{r}-\frac{k}{m}\end{bmatrix}\begin{bmatrix}u_1(t)\\ u_2(t)\end{bmatrix} \quad \text{or} \quad \begin{bmatrix}\ddot{u}_1(t)\\ \ddot{u}_2(t)\end{bmatrix} = \overset{A'}{-}\begin{bmatrix}\frac{g}{r}+\frac{k}{m} & -\frac{k}{m}\\ -\frac{k}{m} & \frac{g}{r}+\frac{k}{m}\end{bmatrix}\begin{bmatrix}u_1(t)\\ u_2(t)\end{bmatrix}$$

$$\hookrightarrow \ddot{U} = -A'U$$

if we "guess" that a solution $U(t)$ will take the form: $U(t) = Xe^{st} : X \in \mathbb{R}^2$

fourier analysis
any analytical function can be made of a linear combination of e^{st}

then: $\frac{d}{dt}[U(t)] = sU(t)$ & $\frac{d}{dt}[U'(t)] = s^2U(t)$

if s is completely imaginary:

$$\Rightarrow \quad Xs^2e^{st} = \overset{M}{A}Xe^{st}$$

$s = \dot{2}a : a \in \mathbb{R}$

$\Rightarrow \boxed{s^2 = -a}$

$e^{st} \to e^{\dot{2}a}$

$$\Rightarrow \quad 0 = e^{st}(AX - Xs^2)$$

$$\Rightarrow \quad 0 = (\underbrace{AX - Xs^2})$$

$$\Rightarrow \quad 0 = X(A - I_2s^2)$$

$$\Rightarrow \quad \det(A - I_2s^2) = 0$$

$(-\dot{2})(-\dot{2})$

$\hookrightarrow \cancel{-}1\cdot\dot{2}\cdot\cancel{-}1\cdot\dot{2}$

$\dot{2}\cdot\dot{2} = -1$

$$\Rightarrow \quad \det\left(\begin{bmatrix}-\frac{g}{r}-\frac{k}{m}-\lambda & \frac{k}{m}\\ \frac{k}{m} & -\frac{g}{r}-\frac{k}{m}-\lambda\end{bmatrix}\right) = 0$$

$$\begin{bmatrix}\frac{g}{r}+\frac{k}{m}-\lambda & -\frac{k}{m}\\ -\frac{k}{m} & \frac{g}{r}+\frac{k}{m}-\lambda\end{bmatrix}$$

STOP Here

$$\Rightarrow \boxed{\left(\frac{g}{r}+\frac{k}{m}+\lambda\right)^2-\frac{k^2}{m^2}=0}$$

$$\Rightarrow \lambda^2+\frac{k^2}{m^2}+\frac{g^2}{r^2}+2\frac{gk}{rm}+\frac{g}{r}\lambda+\frac{k}{m}\lambda-\frac{k^2}{m^2}=0$$

$$\Rightarrow \lambda^2+\lambda\left(\frac{g}{r}+\frac{k}{m}\right)+\left(\frac{g^2}{r^2}+2\frac{gk}{rm}\right)=0$$

if you start with the cos version

$$\left(\lambda-\frac{g}{r}-\frac{k}{m}\right)^2-\frac{k^2}{m^2}=0$$

$$\lambda^2+\frac{g^2}{r^2}+\cancel{\frac{m^2}{k^2}}+2\frac{gk}{rm}-\lambda\frac{g}{r}-\lambda\frac{k}{m}-\cancel{\frac{k^2}{m^2}}=0$$

$$\lambda^2-\lambda\left(\frac{g}{r}+\frac{k}{m}\right)+\frac{g^2}{r^2}+2\frac{gk}{rm}=0$$

if s is entirely complex then s^2 will always be negative.

$$e^{st}X=-s^2e^{st}A'X$$

that makes the two sides of this eq always opposite sign.

$$(A'X-I\lambda) \text{ vs } \boxed{(A'X+I\lambda)}$$

$$\lambda^2+\lambda\left(\frac{g}{r}+\frac{k}{m}\right)+\frac{g^2}{r^2}+2\frac{gk}{rm}$$

$$\frac{-\frac{g}{r}-\frac{k}{m}\pm\sqrt{\frac{g^2}{r^2}+\frac{k^2}{m^2}+2\frac{gk}{rm}-4\left(\frac{g^2}{r^2}+2\frac{gk}{rm}\right)}}{2}$$

$$= \frac{-\frac{g}{r} - \frac{k}{m} \pm \sqrt{\left(-3\frac{g^2}{r^2} + \frac{k^2}{m^2} - 6\frac{gk}{rm}\right)}}{2}$$

$$\left(-3\frac{g^2}{r^2} + \frac{k^2}{m^2} - 6\frac{gk}{rm}\right)$$

$A = \frac{g}{r} \quad B = \frac{k}{m}$

$$-3A^2 - 6AB + B^2$$

$$\sqrt{\left(\frac{g}{r} + \frac{k}{m} + \lambda\right)^2} = \sqrt{\frac{k^2}{m^2}} \quad : \quad \lambda = s^2$$

$$\frac{g}{r} + \frac{k}{m} + \lambda = \pm\frac{k}{m}$$

$$\Rightarrow \lambda = -2\frac{k}{m} - \frac{g}{r} \quad \& \quad \lambda = -\frac{g}{r}$$

$$\Rightarrow \lambda = -\left(2\frac{k}{m} + \frac{g}{r}\right), -\left(\frac{g}{r}\right)$$

$$\lambda = s^2 \Rightarrow s = i\sqrt{2\frac{k}{m} + \frac{g}{r}}, i\sqrt{\frac{g}{r}} \quad \& \quad X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow u = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{i\sqrt{\frac{g}{r}}\, t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{i\sqrt{2\frac{k}{m} + \frac{g}{r}}}$$

$$e^{iWt} =$$

$$e^{i\omega t} =$$

$$\cos(\omega t) = \frac{1}{2} e^{i\omega t} + \frac{1}{2} e^{-i\omega t}$$